

Early psilocybin intervention alleviates behavioral despair and cognitive impairment in stressed Wistar rats

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ABSTRACT

Chronic stress exerts profound effects on mental health, contributing to disorders such as depression, anxiety, and cognitive impairment. This study examines the potential of psilocybin to alleviate behavioral despair and cognitive deficits in a rodent model of chronic stress, focusing on the interplay between the Hypothalamic-Pituitary-Adrenal (HPA) axis and the Endocannabinoid System (ECS). Twenty-two male Wistar rats were divided into control and stress groups. Animals within the stress group were exposed to predator odor and chronic social instability to induce chronic stress, and were either sham treated, or given psilocybin. Behavioral assessments were conducted using the Open Field Test, Sucrose Preference Test, Novel Object Recognition, Elevated Plus Maze, and Forced Swimming Test to evaluate locomotion, anhedonia, memory, anxiety, and behavioral despair, respectively. Blood and brain samples were analyzed for biochemical markers. Results indicated that psilocybin significantly reduced stress-induced behavioral despair and cognitive impairments, likely through ECS-mediated downregulation of the HPA axis. These findings suggest that early intervention with psilocybin has sustained beneficial effects on stress-related behavioral and cognitive disturbances, underscoring its potential as a novel therapeutic approach for stress-related mental health disorders.

1. Introduction

Psilocybin is a classic psychedelic compound derived from “magic mushrooms” (Ziff et al., 2022). In recent years, the therapeutic potential of psilocybin for mental health disorders has drawn significant attention from scientific and commercial aspects (Aday et al., 2023) due to its outstanding safety and efficacy in treating depression and anxiety disorders (Gukasyan et al., 2022). Clinical trials evaluating psilocybin-assisted psychotherapy (PAP) have shown reduced depression scores and sustained effects for patients with major depressive disorder (Gukasyan et al., 2022; Reed and Foldi, 2024; Goodwin et al., 2022) and treatment resistant depression (Goodwin et al., 2022; Rosenblat et al., 2024). Additionally, preliminary results have shown that PAP is effective in reducing drug-seeking and compulsive behaviors in patients with substance use disorders (SUDs) and obsessive-compulsive disorder (OCD) (Bogenschutz et al., 2022; Agin-Liebes et al., 2024; Barrett et al., 2020; Pagni et al., 2024), and may also alleviate intrapersonal disconnection in palliative care patients (Watts et al., 2017; McManus et al.,

2022).

The current understanding on the molecular mechanism of psilocybin focuses on its interaction with the serotonin (5-HT) system, partially due to its chemical similarity to 5-HT (Duan et al., 2024). Among all 5-HT receptors, the 5-HT_{2A} receptor is thought to be the primary target responsible for most of psilocybin’s psychoactive and therapeutic effects (Ling et al., 2022; Inserra et al., 2021), with psilocybin occupying up to 72 % of these receptors at clinically relevant doses (Inserra et al., 2021). However, the mechanism by which psilocybin interacts with the stress response remains controversial.

Chronic stress-induced physiological changes strongly influence mental health conditions and brain chemistry. Increasing evidence suggests that patients with depressive disorders, SUDs and OCD exhibit elevated cortisol levels (Tafet and Nemeroff, 2016; Kirsch and Lippard, 2022; Morgado et al., 2013). Various studies have reported a transient increase in the circulating levels of adrenocorticotrophic hormone (ACTH) and corticosteroids (CORTs) (de Veen et al., 2017; Mason et al., 2023; Jones et al., 2023), attributed to serotonin 2 A (5-HT_{2A}) receptor-

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mediated activation of the HPA-axis (Inserra et al., 2021). A recent clinical trial has reported that participants in the psilocybin-stress group demonstrated a lower cortisol increase than those in the placebo group 7 days post administration, suggesting a persistently blunted effect of psilocybin on the cortisol response (Mason et al., 2023). However, it remains unclear how psilocybin affects chronic stress, and more evidence is needed to confirm the impact of psilocybin on the physiological response of stress.

The objective of this study is to investigate the potential effects of early intervention with psilocybin on stress response, behavioral modification, and neurochemical changes using a rodent model. Our results demonstrate that early intervention with psilocybin 1) reduces the likelihood of developing mental health disorders triggered by chronic stress; 2) has long-lasting effects even in the presence of ongoing chronic stress; and 3) affects physiological responses and brain chemistry. This study uniquely leverages early intervention with psilocybin, rather than treatment, to enhance both cost-effectiveness and safety.

2. Materials and methods

2.1. Animals

All procedures involving animals strictly adhered to the Guidelines of the Canadian Council on Animal Care. All testing and treatment procedures were reviewed and approved by the Animal Care and Use Committees of the Health Sciences Laboratory Animal Services at the University of Alberta. Twenty-two male WIS rats (≈ 500 g), aged 60 days, were purchased from Charles River Laboratories. All rats were habituated in the animal facility, under a 12-h light/dark cycle, a consistent temperature of 22 °C, and humidity between 50 and 55 % (Wang et al., 2023). Rats were housed in pairs with food and water *ad libitum* for the duration of the experiment (45 days), except where described, and all procedures were conducted during the light cycle (06:00–18:00), except the second POE (18:00–6:00).

2.2. Experimental design

Following 7 days of habituating the animals to handling, the animals were randomly divided into three groups: one control group (CTL, $N = 6$) and two stress groups (SIS, $N = 16$). Due to the potential variability of response to stress and psilocybin, two additional rats were included in each stress group, bringing the total to 8 animals per stress group. The SIS groups were divided into two subgroups: the sham-treatment group (SIS-Sham) and the psilocybin-treatment group (SIS-PSI). As previously described (Wang et al., 2023), on the first day of the experiment, animals were exposed to the first predator odor exposure (POE). Animals in

stress groups were subjected to chronic social instability stress (SIS) exposure for 10 days. On Day 12, a second POE was conducted to strengthen the stress stimulation, which was followed by additional SIS exposure for 18 days (Wang et al., 2023). In contrast, CTL animals were left undisturbed during the SIS exposure periods but underwent both POEs using distilled water (Wang et al., 2023). Interventions were administered intraperitoneally (IP) on all animals one hour after their POE experience (Ritov and Richter-Levin, 2017; Xue et al., 2019). Between days 33–39, all animals underwent behavioral tests and were then euthanized on day 40, when blood samples and brain tissues were collected for further analysis (Fig. 1).

2.3. Predator odor exposure (POE)

Adapted from our previous work (Wang et al., 2023), briefly, 2,5-dihydro-2,4,5-trimethylthiazoline (TMT, 97 % purity, SRQ Bio), a chemical compound extracted from red fox urine (Chan et al., 2021), was used to stimulate the presence of a predator and to trigger innate fear and stress responses (Tyler et al., 2020). During the 1-h POE, rats were placed individually into a clean cage containing a 50 ml conical centrifuge tube (14-432-22, Fisher Scientific). The tube contained a piece of filter paper moistened with 5ul of TMT (or 5ul of distilled water for the CTL group). The lid of the tube and tube body were drilled with small holes allowing the rats to smell the odor while preventing the rats from directly interacting with the filter paper. The first POE was performed during the light cycle (06:00–18:00) on Day 1, while the second POE occurred during the night cycle (18:00–06:00) on Day 12. Repeated POEs reinforced the animals' original stress response through recurrent experience (Ojo et al., 2014). The second POE was conducted during the night cycle, strategically disrupting the animals' anticipatory mechanisms, thus heightening the stressor's unpredictability and intensifying the overall stress response (Ojo et al., 2014).

2.4. Drug delivery

1.0 mg/kg psilocybin solution was prepared under sterile conditions by dissolving psilocybin (99.7 % purity, Psygen Labs Inc) in sterile saline (0.9 %). The psilocybin solution was freshly prepared on the day of the POE, vortexed well before loading into the syringe, then immediately IP injected into each rat to guarantee administration of a homogeneous solution (Huang et al., 2022). Animals in CTL and SIS-Sham groups were IP injected with sterile saline (0.9 %, vehicle) and animals in SIS-PSI group were injected with psilocybin (1.0 mg/kg) at a volume of 1 ml/kg of mass (Huang et al., 2022; Malikowska-Racia et al., 2023). Following the POE, each rat was carefully groomed with a clean, dry towel to remove any residual feces or odors from the POE apparatus

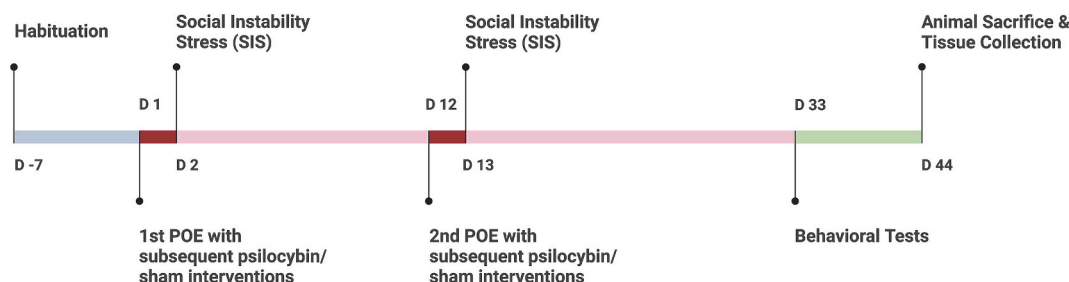


Fig. 1. Experimental timeline. All animals were habituated for 7 days prior to the experiment. Animals in all groups were subjected to two predator odor exposure (POE) events (SIS groups were exposed to TMT while the CTL group was exposed to distilled water) on day 1 (6:00–18:00) and day 12 (18:00–6:00). Rats in the stress groups were injected intraperitoneally with either sterile saline (0.9% NaCl) (SIS-Sham) or 1.0mg/kg psilocybin (SIS-PSI) one hour after the POE. Between days (Ds) 2–11 and 13–32, rats in stress groups (SIS-Sham and SIS-PSI) were subjected to chronic SIS, while rats in the control group (CTL) were left undisturbed. A series of behavioral tests were conducted between Ds 33–43 to evaluate the general activity levels (D 33), anhedonia-like behavior (Ds 34–38), recognition memory (Ds 39–40), anxiety-like behavior (D 41), and behavioral despair (Ds 42–43), followed by the collection of cardiac blood and brain tissue on D 44. TMT: dihydro-2,4,5-trimethylthiazoline; OFT: open field test; SPT, sucrose preference test; NOR: novel object recognition; EPM: elevated plus maze; FST: forced swim test. Figure was created with Biorender.com.

before being returned to its home cage. Interventions (sham or psilocybin) were administered within one hour after the POE by gently restraining the animal with a rodent surgical drape.

2.5. Social instability stress (SIS)

The social instability stress (SIS) paradigm was developed based on previous studies (Koert et al., 2021; Planchez et al., 2019) (Wang et al., 2023). The animals in the CTL group were left undisturbed. Cage partners in the SIS groups were changed daily between days 2–11 and days 13–32. To prevent prolonged stays in the same cage and reduce the likelihood of a rat becoming a dominant and aggressive resident animal, a strict rotation was employed. In a 7-day rotation, within same SIS subgroup, each rat was paired for one 24-h period with each of the other 7 rats in the same subgroup. Each rat spent a maximum of two days in the same cage, whereby the rat that had been introduced into a new cage remained in the same cage on the following day when a new partner was introduced. On the third day, the “resident” rat was introduced into a new cage as the novel partner.

2.6. Behavioral tests

A series of behavioral tests were conducted to assess general activity levels, anhedonia-like behavior, recognition memory, anxiety-like behavior, and behavioral despair. Prior to each test, the animals were transported to the testing room and allowed to acclimate for a minimum of 30 min. All testing apparatus were sanitized with disinfectant prior to and after each animal. The behavioral measurements were documented and analyzed using the ANY-maze software (Version 7.20). Only one behavioral test was conducted per day, with a minimum of a 24-h interval between tasks. The behavioral tests were performed in the following order: Open field test (OFT), sucrose preference test (SPT), novel objective recognition (NOR), elevated plus maze (EPM), and forced swimming test (FST).

2.6.1. Open field tests (OFT)

The OFT was utilized to assess exploratory activity and locomotion according to previous protocols (De Gregorio et al., 2022). The OFT apparatus, comprised of a black PVC plastic box with dimensions of 48*48*50 cm (L*W*H), was virtually divided into 16 identical squares using the ANY-maze software. The outer zone (consisting of 12 squares) was designated as the peripheral zone, while the central area consisted of 4 squares was labeled as the centre zone. Rats were introduced into the OFT apparatus and allowed to explore freely for 15 min. Distance travelled (in cm) and immobile time (in sec) were recorded and evaluated to measure exploratory activity and locomotion.

2.6.2. Sucrose preference test (SPT)

The SPT was employed to evaluate anhedonia-like behavior in rodents (Liu et al., 2018). We adapted the 6-day SPT protocol comprised of four periods (acclimatization, training, deprivation and testing), from previous studies (De Gregorio et al., 2022; Liu et al., 2018; He et al., 2020). During the acclimatization period (day 1–2), each animal was individually housed in a cage containing two bottles of drinking water and *ad libitum* access to food chow to acclimatize the animals to the option of two bottles (Liu et al., 2018). For training, animals were given two bottles of a 1 % (w/v) sucrose solution (day 3), followed by replacing only one bottle with drinking water and the other remaining filled with 1 % (w/v) sucrose solution (day 4) (He et al., 2020). The position of the two bottles changed once per day to avoid conditioned place preference learning (De Gregorio et al., 2022). On day 5, all animals were deprived of food and water for 12 h overnight (18:00–06:00). Immediately after deprivation (day 6), all animals were provided with a pre-weighed bottle of drinking water and a pre-weighed bottle with a 1 % (w/v) sucrose solution. All animals were allowed free access to both bottles for a total of 8 h (during the light cycle), exchanging the bottle positions every 2 h

to prevent conditioned place preference learning (De Gregorio et al., 2022). The final weight of bottles was recorded to calculate the sucrose preference (%), which was determined by sucrose consumption (in g) / total consumption (in g) * 100 (De Gregorio et al., 2022).

2.6.3. Novel objective recognition (NOR)

The NOR protocol was preformed in the open field arena and adapted from our previous study (Wang et al., 2023). During training (day 1), two objects were placed in the box at adjacent corners (9 cm from the wall) before introducing the rats. Each rat was initially placed facing away from both objects and allowed to freely explore the arena and objects. On the test day (day 2), a novel object was placed in a third adjacent corner (9 cm from the walls) (Wang et al., 2023). Each rat, initially facing away from all objects, was placed back in the apparatus and allowed to freely explore the arena for 5 min. Time spent interacting with each object was recorded to evaluate recognition memory, which is based on the innate preference for novelty in rodents as previously indicated (Chesworth et al., 2018; Akkerman et al., 2012). The recognition index (RI), representing the percentage of time spent with the novel object, was used to measure novelty preference and memory retention. RI was calculated by dividing the time spent exploring the novel object (T_N) by the total time exploring both the novel and familiar objects (T_F) ($RI = T_N / (T_N + T_F)$) (Antunes and Biala, 2012).

2.6.4. Elevated plus maze (EPM)

The EPM apparatus consisted of two open arms with dimensions of 50*10 cm (L*W) and two closed arms of equal size with 45 cm high walls. The entire apparatus was elevated 55 cm off the floor. Rats were positioned at the intersection of the four arms, facing an open arm, and allowed to freely explore the arena for a duration of 5 min. Time spent in closed and open arms was recorded to assess anxiety-like behavior. In addition, we measured the time spent engaged in risk assessment behavior to improve the predictive validity of the EPM (Correa-Netto et al., 2017; Walf and Frye, 2007). Risk assessment behavior was defined as a stretched-attend posture, wherein the animal elongates its neck and shoulders towards both the open and closed arms, followed by a retraction to the original posture (Correa-Netto et al., 2017; Walf and Frye, 2007).

2.6.5. Forced swimming test (FST)

The FST was performed in Plexiglas cylinders (internal D*H: 20*50 cm) filled with 23–25 °C water, ensuring that the water depth prevented the rat from touching the container bottom with its hind legs or tail. Adapted from previous studies (Wang et al., 2023; Yankelevitch-Yahav et al., 2015; Slattery and Cryan, 2012), the FST protocol spanned 2 days and was comprised of one day of habituation and one day of testing. On the first day, each rat was individually introduced into the warm water-filled cylinder for a 15-min habituation. Rats were removed from the water, dried with clean towels and transferred to a clean cage on a heating pad until fully dry before returning them to their home cages. On the second day, each rat was re-introduced to the water-filled cylinder for testing with a duration of 10 min. Following a 2-min acclimatization, the duration of time each rat spent inactive and active behaviors was recorded for 8 min. The immobile time, reflecting inactive component of behavior, was documented when a rat makes only necessary movements to keep its head above the water, remain floating, or turn slightly (De Gregorio et al., 2022). As per previous studies (Bogdanova et al., 2013; Fenton et al., 2015), we further classified active behaviors into swimming and climbing. Time spent climbing was recorded when animal vertically moved against walls, while swimming time was identified when animal presented horizontal movement across the water surface (Bogdanova et al., 2013).

2.7. Blood and brain sample collection

Animals were deeply anesthetized with isoflurane (5 %, carried by

oxygen) and positioned in dorsal recumbency. Cardiac blood samples were collected with 20-gauge 1.5" needles (305,175, Becton Dickinson) into blood collection tubes (0268393 A, Fisher Scientific). After clotting for 30 min at room temperature, the blood samples were centrifuged for 10 min at 4 °C and 1000 ×g. Serum was immediately aliquoted into 1.5 ml microcentrifuge tubes (05–408-129, Fisher Scientific) until further use. Following cardiac blood collection, all rats were perfused intracardially with 1 × PBS (BP3994, Fisher Scientific). Six out of eight brain tissues were collected and dissected into the left and right hemispheres. The right hemisphere was further dissected into the prefrontal cortex (PFC), amygdala (AMG), hippocampus (HIP), and hypothalamus (HYP). The left hemisphere was post-fixed in 4 % PFA for 16 h and then transferred to 1 × PBS for future use. Additionally, two rats from each group were intracardially pre-fixed with 4 % paraformaldehyde (PFA, A11313.22, Thermo Fisher Scientific). The brain tissues were collected in their entirety without dissection, followed by a 16-h post-fixation using 4 % PFA, and then transferred to 1 × PBS for future use.

2.8. Luminescence assay

To evaluate the levels of circulated BDNF and representative hormones affected by stress, including ACTH and CORT, serum samples were 2-fold diluted in 1 × PBS and subsequently analyzed using the Luminescence assay analysis (RSH-02-205 and RPT-04-212, Eve Technologies). The remaining serum samples were frozen at –80 °C until future use.

2.9. Enzyme-linked immunosorbent assay (ELISA)

ELISA assays were performed to measure the 2-AG levels in various brain regions. Dissected brain samples (the PFC, AMG, HIP, and HYP) were weighted and then lysed in RIPA buffer supplemented with protease inhibitor (A32963, Thermo Scientific) and phosphate inhibitor (A32957, Thermo Scientific). The tissue samples were homogenized using the tissue homogenizer (PRO Scientific Bio-Gen PRO200, Cole-Parmer) and incubated on ice for 20 min. The supernatants were acquired by centrifuging the brain tissue lysates at 4 °C, 1200 ×g for 30 min then frozen at –80 °C. Evaluation of 2-AG level was conducted using ELISA detection kits (MBS1600322, MyBioSource, Inc.).

2.10. Western blot (WB)

WB was utilized to quantify the expression levels of cannabinoid receptor type I (CB₁R), BDNF, receptor tyrosine kinase B (TrkB), extracellular signal-regulated kinase (ERK), protein kinase B (Akt), and mammalian target of rapamycin (mTOR). Protein lysates, prepared as described for ELISA analysis, were quantified using the Bradford protein assay kit (5,000,002, Bio-Rad) and measured with the FLUOstar Omega Microplate Reader (BMG). Protein extracts (20 µg/lane) were loaded and separated by hand-cast 10 % Sodium Dodecyl Sulfate Polyacrylamide Gel Electrophoresis (SDS-PAGE) gels. Proteins were then transferred onto a 0.45 µm nitrocellulose membrane (1,620,115, Bio-Rad). After blocking with 5 % bovine serum albumin (BSA) prepared in 1 × tris-buffered saline (TBS), membranes were incubated with primary antibodies in 1 × TBS with 0.1 % Tween-20 (TBST) overnight at 4 °C. Membranes were then washed in 1 × TBST three times, 10 min each, and incubated with secondary antibodies at room temperature for 1 h. Following three washes in TBST, protein bands on membranes were visualized and quantified using Image Studio Lite (LI-COR Biosciences, Lincoln, NE Image Studio software Ver 5.2). Membranes were re-probed with GAPDH, employed as a loading control, to ensure equal sample loading, with all band intensities normalized to GAPDH levels.

The primary antibodies used: rabbit anti-CB₁R (1:500; ab23603, Abcam), rabbit anti-BDNF (1: 1000; ab108319, Abcam), rabbit anti-TrkB (1:1000; 4603S, Cell Signaling), rabbit anti-ERK1/2 (1:1000; 9102S, Cell Signaling), rabbit anti-p-ERK1/2 (1: 1000; 4376S, Cell

Signaling), mouse anti-Akt (1:1000; 2920S, Cell Signaling), rabbit anti-p-Akt (1:1000; 4060S, Cell Signaling), rabbit anti-mTOR (1:1000; 2983S, Cell Signaling), rabbit anti-p-mTOR (1:1000; 5536S, Cell Signaling), and mouse anti-GAPDH (1:1000; 97166S, Cell Signaling). The following secondary antibodies were used in a 1:1000 dilution: goat anti-rabbit-IR800 (926–32,211, LI-COR Biosciences) and goat anti-mouse-IR680 (926–68,070, LI-COR Biosciences).

2.11. Statistical analyses

The data were presented as mean ± SEM. PRISM software (Ver 10.2.3, GraphPad Software) was employed to generate graphs and conduct statistical analysis. Comparisons among CTL group, SIS-Sham subgroup, and SIS-PSI subgroup were evaluated by one-way ANOVA followed by Tukey's post-hoc. All tests between SIS-Sham and SIS-PSI subgroups were evaluated by performing *t*-tests. Statistical significance was designated at a *p*-value <0.05.

3. Results

3.1. Chronic stress-induced behavioral changes in Wistar rats are ameliorated by early psilocybin intervention

In order to assess whether 1.0 mg/kg psilocybin was able to ameliorate chronic stress induced behavioral changes in Wistar rats, a battery of behavior assays were utilized following the chronic stress protocol. According to our OFT results (Fig. 2A), SIS groups were less active than the CTL group, travelling less distance (Fig. 2A, $F(2,16) = 27.07$, $p < 0.0001$; Tukey's multiple comparisons test, $p < 0.0001$, $p = 0.0001$) and spending more time immobile (Fig. 2B, $F(2,16) = 9.684$, $p = 0.0018$; Tukey's multiple comparisons test, $p = 0.0013$) than the CTL group. This inactivity phenotype was slightly, though not significantly reversed, in the psilocybin treated group (Fig. 2A, $p = 0.2060$; Fig. 2B, $p = 0.0875$).

In the EPM, a significant difference was observed in the amount of time the animals spent in the open arms ($F(2, 16) = 3.881$, $p = 0.0423$) and the number of entries into the open arms ($F(2,16) = 8.100$, $p = 0.0037$). The SIS-Sham group spent significantly less time in the open arms (Fig. 2C, $p = 0.0336$) and entered the open arms significantly less frequently (Fig. 2E, $p = 0.0059$) compared to the CTL group. Early psilocybin intervention significantly increased the number of entries into the open arms (Fig. 2E, $p = 0.0148$) compared to the SIS-Sham group, but while the amount of time spent in the open arms increased in the SIS-PSI group compared to the SIS-SHAM group, it was not significant (Fig. 2C). No differences were observed in the amount of time spent in the closed arms among the three conditions (Fig. 2D, $F(2,16) = 0.3328$, $p = 0.7218$). In addition, risk assessment behaviors were assessed, whereby the time spent in a stretch-attend posture was measured. One-way ANOVA results shown significant differences in the time spent in risks assessment behaviors among three conditions ($F(2, 16) = 10.10$, $P = 0.0015$), whereby, Tukey's post-hoc test demonstrated that the SIS-Sham subgroup spent more time in risk assessment behavior than the CTL group (Fig. 2F, $p = 0.0012$), while the SIS-PSI subgroup spent less time assessing risk than the SIS-Sham group (Fig. 2F, $p = 0.0401$).

To evaluate despair-like behaviors, the FST was utilized and revealed significant differences among the three different conditions ($F(2, 16) = 10.48$, $p = 0.0012$). The SIS-Sham group spent less time moving compared to the CTL group (Fig. 2G, $p = 0.0009$), considered to be a feature of despair-like behaviors in rats (Yankelevitch-Yahav et al., 2015), which was alleviated upon early psilocybin intervention (Fig. 2G, $p = 0.0444$). Interestingly, time spent swimming (Fig. 2I, *t*-test, $p = 0.0379$) as opposed to time spent climbing (Fig. 2H) was significantly increased in psilocybin treated rats (SIS-PSI) compared to the sham treated rats (SIS-Sham).

Anhedonia-like behaviors were also evaluated in the chronic stress

exposed rats, using the SPT, comparing the consumption of 1 % (w/v) sucrose solution *versus* normal drinking water. The psilocybin treated group demonstrated a higher sucrose preference (%) than the sham treated rats (Fig. 2J, t-test, $p = 0.0495$).

To test if psilocybin intervention was able to alter recognition memory, we preformed the NOR test, and observed significant differences in RI were observed among the three different conditions using the one-way ANOVA test ($F(2, 15) = 13.04$, $p = 0.0005$). The sham treated group demonstrated a significantly lower RI than the CTL group (Fig. 2K, $p = 0.0004$), while the SIS-PSI group had a significantly higher RI than the SIS-Sham group (Fig. 2K, $p = 0.0250$).

3.2. Early intervention psilocybin downregulates stress-induced HPA-axis hyperactivity

The hypothalamic-pituitary-adrenal (HPA) axis maintains homeostasis through intricate hormonal modulation, regulating stress hormone release and physiological responses to perceived threats (Sheng et al., 2020; Oster et al., 2017). Upon stress perception, the HYP secretes corticotrophin-releasing hormone (CRH) (Herman et al., 2016; Roberts and Karatsoreos, 2021), prompting the pituitary gland to release adrenocorticotrophic hormone (ACTH) (Sheng et al., 2020), which stimulates the adrenal cortex to produce corticosteroids (CORTs) (Joëls et al., 2018; Ramamoorthy and Cidlowski, 2016). The levels of stress hormones, ACTH and CORT, in the blood serum were measured using the Luminex assay. Significant differences were observed in the level of ACTH among the three conditions ($F(2, 16) = 9.129$, $p = 0.0023$), whereby the sham-treated group (SIS-Sham) exhibited significantly higher levels of ACTH compared to the CTL group (Tukey's post-hoc test (Fig. 3A, $p = 0.0039$). While psilocybin intervention (SIS-PSI) significantly reduced ACTH levels compared to the Sham group (Fig. 3A, $p = 0.0093$). Similarly, CORT levels also exhibited significant differences ($F(2, 16) = 10.11$, $p =$

0.0015), revealing a similar trend as observed in ACTH levels. The SIS-Sham subgroup exhibited a significantly higher level of CORT compared to the CTL group (Fig. 3B, $p = 0.0011$), while the SIS-PSI subgroup had a significantly decreased CORT level than the SIS-Sham subgroup (Fig. 3B, $p = 0.0418$).

3.3. Psilocybin restores BDNF levels and upregulates TrkB expression

BDNF is a neurotrophin and neurotransmitter primarily synthesized in the hippocampus (HIP) (García-Suárez et al., 2020) and highly expressed in the HYP (Noble et al., 2011). A deficit in BDNF has been linked to disrupted brain plasticity and the pathogenesis of mental health conditions (Colucci-D'Amato et al., 2020). The levels of BDNF were measured in blood serum using the Luminex assay and four representative brain regions using Western blot analysis. Significant differences in the circulating levels of BDNF among the three different conditions were observed (One-way ANOVA, $F(2, 16) = 6.470$, $p = 0.0087$). The SIS-Sham subgroup exhibited a significantly lower circulating BDNF level than the CTL group (Fig. 4A, $p = 0.0319$), while the SIS-PSI subgroup had significantly elevated circulating levels of BDNF compared to the SIS-Sham group (Fig. 4A, $p = 0.0124$). No significant differences were observed between the SIS-PSI subgroup and CTL group (Fig. 4A).

Within the brain of the SIS groups, Western blot data (Fig. 4B) confirmed that psilocybin intervention significantly increased the level of BDNF in the PFC, AMG and HYP, though not in the HIP, compared to the sham treated group (Fig. 4C, $p = 0.0229$, $p = 0.0339$, and $p = 0.0229$). Interestingly, the receptor for BDNF, TrkB, expression was upregulated in all four brain regions (PFC, AMG, HYP and HIP) in the SIS-PSI group compared to the SIS-Sham group (Fig. 4D, $p = 0.0076$, $p = 0.0120$, $p = 0.0186$, $p = 0.0138$).

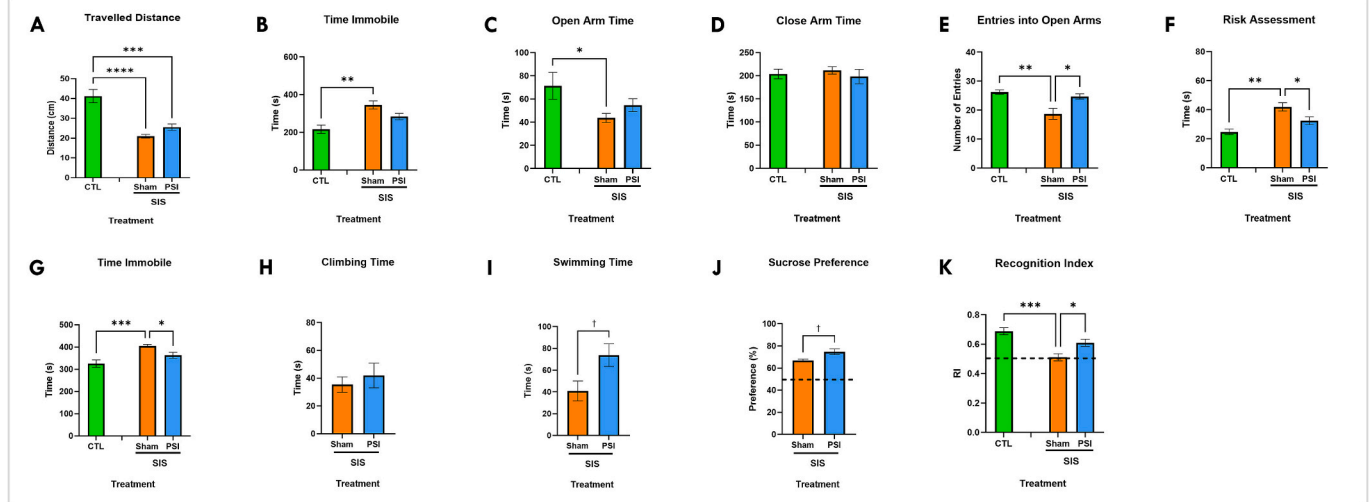


Fig. 2. Chronic stress induces behavioral changes in Wistar rats, which is ameliorated by early intervention with psilocybin. (A) Rats in the social instability stress (SIS) groups travelled less distance than rats in the control (CTL) groups during the open field test. A slight, but not significant, increase in distance travelled was observed in psilocybin treated rats (SIS-PSI) as opposed to sham treated rats (SIS-Sham). (B) SIS-Sham rats presented a greater immobile time (seconds, s) than CTL rats, a slight but not significant decrease in immobile time was observed between SIS-PSI rats and SIS-Sham rats. (C) SIS-Sham rats spent less time (sec) in the open arms of the elevated plus maze (EPM) compared to the CTL rats, while a slight but not significant increase in time spent in the open arms was observed in the SIS-PSI rats compared to the SIS-Sham rats. (D) No significant differences were observed in time (sec) spent in the closed arms of the EPM among the three groups. (E) SIS-Sham rats made less entries into open arms of the EPM than CTL rats, while the SIS-PSI rats made more entries than the SIS-Sham rats. (F) SIS-Sham rats spent more time (sec) in risk assessment behavior than CTL rats, while the SIS-PSI rats spent less time (sec) assessing risk than the SIS-Sham rats. (G) SIS-Sham rats showed a greater immobility than the CTL rats in the forced swim test (FST), while the SIS-PSI rats spent less time immobile than SIS-Sham rats. (H) SIS-PSI rats exhibited a slight but insignificant increase in the climbing time (sec) in the FST, (I) while they had a significant increase in the swimming time (sec) compared to the SIS-Sham group. (J) SIS-PSI rats demonstrated a higher sucrose preference (%) than the SIS-Sham rats. (K) SIS-Sham rats showed a significantly lower recognition index (RI) than the CTL rats in the novel object recognition test, while the SIS-PSI rats had a significantly higher RI than the SIS-Sham rats. The dotted line indicates the chance performance value (RI = 0.5). For A-G and K: CTL $n = 5$, SIS-Sham $n = 7$, SIS-PSI $n = 7$. Data are expressed as mean $\pm$ SEM (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$). One-way ANOVA followed by Tukey's post hoc test. For H-J: SIS-Sham $n = 7$, SIS-PSI $n = 7$. Data are expressed as mean $\pm$ SEM († $p < 0.05$). t-test.

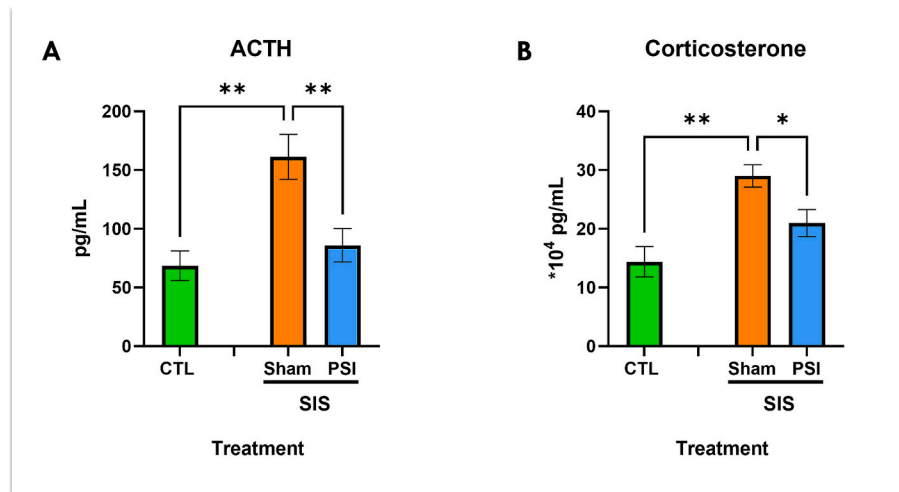


Fig. 3. Chronic stress increases levels of stress hormones, while psilocybin intervention lowers the levels of these hormones in the blood serum. (A) Using the Luminex assay, stress hormone levels in the blood serum were measured. Sham treated stressed rats (SIS-Sham) exhibited a significantly higher level of ACTH compared to rats in the control (CTL) group, while the psilocybin treated stressed rats (SIS-PSI) had a significantly decreased ACTH level than the SIS-Sham rats. (B) SIS-Sham rats exhibited a significantly higher level of CORT compared to the CTL rats, while the SIS-PSI rats had a significantly decreased CORT level than the SIS-Sham rats. No significant differences were observed between the SIS-PSI and CTL rats in both measurements. CTL $n = 5$, SIS-Sham $n = 7$, SIS-PSI $n = 7$. Data are expressed as mean $\pm$ SEM (* $p < 0.05$, ** $p < 0.01$). One-way ANOVA followed by Tukey's post hoc test.

3.4. Psilocybin impacts downstream activation of ERK, Akt and mTOR signaling pathways to different levels in different brain regions

Western blot analysis was conducted in all four brain regions to assess the phosphorylation of the TrkB downstream signaling pathways (Fig. 4B), including ERK, Akt, and mTOR, in the SIS groups (Fig. 4E, F, G). Psilocybin intervention significantly enhanced ERK phosphorylation in the PFC, HIP, and HYP (Fig. 4E, $p = 0.0386$, $p = 0.0114$, $p = 0.0440$), while increased Akt phosphorylation was observed only in the AMG (Fig. 4F, $p = 0.0052$), when compared to the sham treatment group. Moreover, significantly increased mTOR phosphorylation was observed in the PFC and HYP (Fig. 4G, $p = 0.0072$, $p = 0.0361$), while a slight but insignificant enhancement of mTOR phosphorylation was recorded in the AMG and HIP in the SIS-PSI subgroup when compared to the SIS-Sham subgroup (Fig. 4G).

3.5. Psilocybin dampens the stress-induced eCB upregulation and promotes CB₁R expression

Endocannabinoids (eCBs), along with cannabinoid receptors (CBRs) and associated enzymes, make up the neuromodulatory endocannabinoid system (ECS) (Lu and Mackie, 2021), which dampens the excitatory synaptic drive, CRH neuronal activity, and downregulates the activation of the HPA-axis (Wang et al., 2023; Hill et al., 2010). The two most well-studied eCBs are anandamide (AEA) and 2-arachidonoyl glycerol (2-AG) (Reggio, 2010). Previous studies have demonstrated consistent and sustained changes of 2-AG levels (Danan et al., 2021; Morena et al., 2016), while AEA levels exhibit immediate and dynamic alteration in response to stress (Mayo et al., 2020; Dlugos et al., 2012; Hill et al., 2009). Additionally, the 2-AG level is approximately 1000 times higher than AEA in the brain (Zou and Kumar, 2018). Therefore, we specifically focused on the levels of 2-AG, instead of the AEA, to take these advantages and investigate the long-lasting impacts of psilocybin on the eCBs. In the HYP, the center of HPA-axis regulation and the eCB synthesis, the level of 2-AG was quantified using ELISA, and significant differences were found among the three conditions (Fig. 5A, $F(2,10) = 6.852$, $p = 0.0134$). Chronic stress (SIS-Sham) significantly increased 2-AG levels compared to the CTL group (Fig. 5A, $p = 0.0280$), while the psilocybin intervention (SIS-PSI) significantly reduced 2-AG levels compared to the sham treated group (SIS-Sham) (Fig. 5A, $p = 0.0247$). No significant

differences were observed between the SIS-PSI subgroup and CTL group (Fig. 5A). Within the hippocampus (HIP), the level of 2-AG was significantly lower in the psilocybin intervention group compared to the sham treated group (Fig. 5B, t -test, $p = 0.0317$). Moreover, the level of 2-AG receptor, CB₁R, in the AMG, HIP and HYP, was significantly higher in the psilocybin intervention group compared to the SIS-Sham group (Fig. 5C,D, t -test, $p = 0.0148$, $p = 0.0295$, $p = 0.0236$), though only insignificantly increased in the PFC (Fig. 5C,D, $p = 0.209230$).

4. Discussion

Prior studies have demonstrated psilocybin's rapid and sustained benefits in alleviating mood and cognitive deficits, primarily in models where psilocybin was administered under stress-free conditions or after stressor resolution (Kolasa et al., 2024; Hesselgrave et al., 2021; Sekssaoui et al., 2024; Zhao et al., 2024). This study adopts a novel approach by evaluating psilocybin's efficacy when administered immediately after an acute traumatic event, predator odor exposure (POE), and during ongoing chronic stress, modeled by social instability stress (SIS). This design mirrors real-life scenarios where therapeutic interventions often occur amidst persistent stressors. By simulating these conditions, we aim to provide a deeper understanding of psilocybin's potential as early intervention during the occurrence of traumatic events and persistent stress, addressing a critical gap in the current body of research.

Our model of psychosocial stress in combination with predator odor exposure significantly altered the behavior of WIS rats. These behavioral alterations were alleviated to a certain extent by an early intervention with low-dose psilocybin, aligning with previous findings that support using psilocybin as a treatment method immediately after the removal of stress (Hesselgrave et al., 2021; Sekssaoui et al., 2024). Psilocybin is thought to act through serotonin (5-HT) targeting (Foster et al., 2021). Locomotor activity and open arm exploration in the EPM in rodents is closely correlated to levels of neurotransmitters like dopamine (DA) and norepinephrine (NE) (Schank et al., 2008; Rodgers et al., 1994). Notably, even though not significant, the psilocybin-induced improvement on the general activity, in addition to the significant increase of open arm exploration, hints that psilocybin may have DA and NE targeting properties in addition to its well-known serotonin (5-HT) targeting characteristics (Madsen et al., 2019; Grandjean et al., 2021).

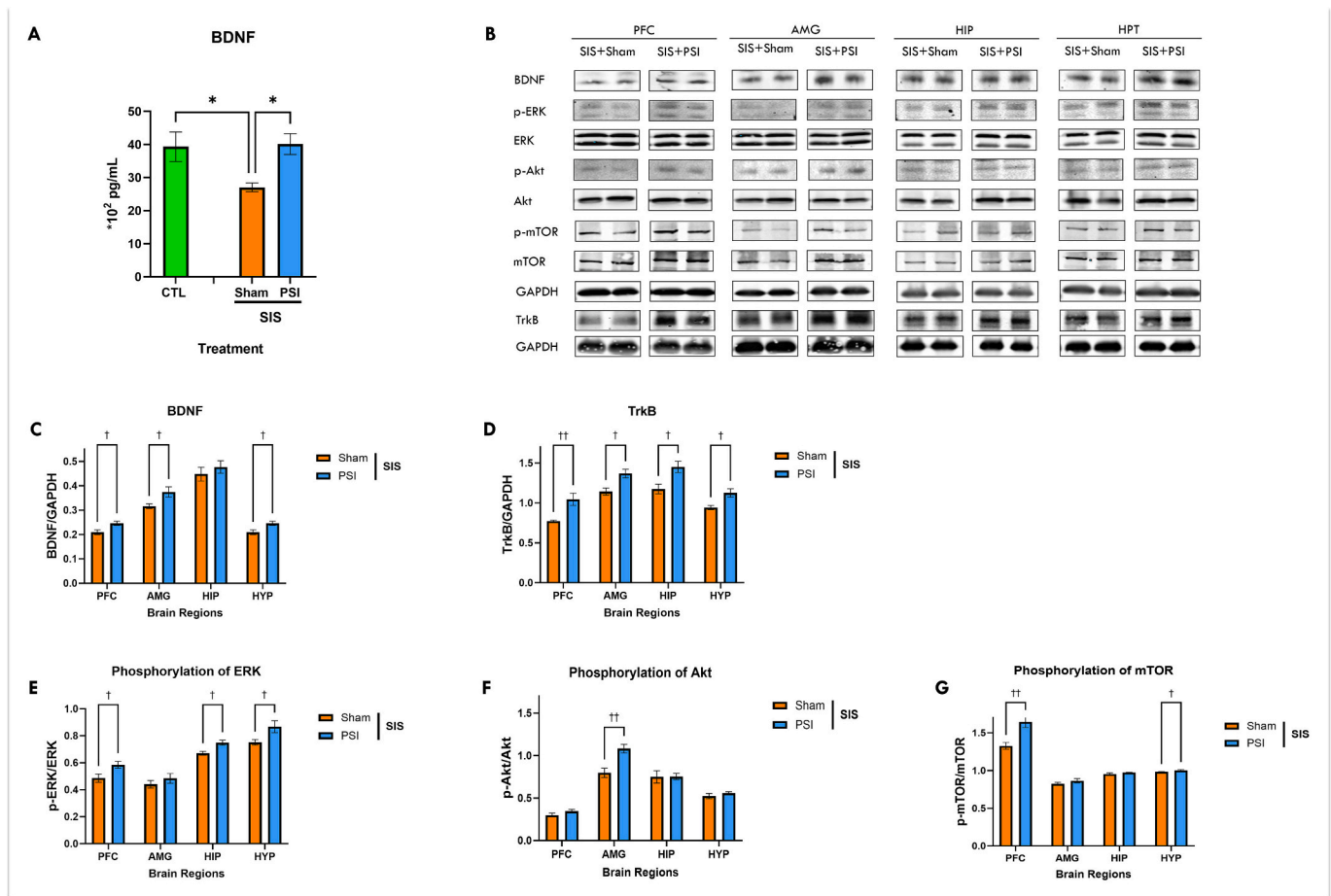


Fig. 4. Chronic stress and early intervention with psilocybin alter BDNF and TrkB levels and differentially activate ERK, Akt, and mTOR signaling pathways across brain regions. (A) The Luminex assay was used to measure circulating BDNF levels in the blood serum. Sham-treated stressed rats (SIS-Sham) exhibited a significantly lower circulating BDNF level than rats in the control (CTL) group, while psilocybin-treated stressed rats (SIS-PSI) had significantly elevated circulating BDNF level than the SIS-Sham rats. CTL $n = 5$, SIS-Sham $n = 7$, SIS-PSI $n = 7$. Data are expressed as mean $\pm$ SEM (* $p < 0.05$). One-way ANOVA followed by Tukey's post hoc test. (B) Representative western blot images probing levels of BDNF, TrkB, p-ERK/ERK, p-Akt/Akt, and p-mTOR/mTOR, in the prefrontal cortex (PFC), amygdala (AMG), hippocampus (HIP), and hypothalamus (HYP) of SIS rats, with GAPDH as a loading control. (C) Quantification of BDNF levels in all four brain regions: SIS-PSI rats exhibited a significant higher BDNF level than the SIS-Sham rats in the PFC, AMG, and HYP. A slight but insignificant increase in the BDNF level in the HIP was observed in the SIS-PSI rats when compared to the SIS-Sham rats. (D) Quantification of TrkB levels in all four brain regions: SIS-PSI rats presented a remarkable upregulation of TrkB expression compared to the SIS-Sham rats. (E) Quantification of p-ERK/ERK levels in all four brain regions: SIS-PSI rats exhibited a significant enhanced ERK phosphorylation than the SIS-Sham rats in the PFC, HIP, and HYP, while a slight but insignificant enhancement was observed in the AMG compared to the SIS-Sham rats. (F) Quantification of p-Akt/Akt levels in all four brain regions: SIS-PSI rats exhibited a significant enhanced Akt phosphorylation than the SIS-Sham rats exclusively in the AMG, while slight but insignificant enhancements have been noticed in the rest three brain regions compared to the SIS-Sham rats. (G) Quantification of p-mTOR/mTOR levels in all four brain regions: SIS-PSI rats exhibited a significant enhanced mTOR phosphorylation in the PFC and HYP, while a slight but insignificant enhancement has been recorded in the AMG and HIP compared to the SIS-Sham rats. SIS-Sham $n = 5$, SIS-PSI $n = 5$. Data are expressed as mean $\pm$ SEM ($\dagger p < 0.05$, $\dagger\dagger p < 0.01$). Multiple unpaired t-test.

Previous research suggests that antidepressants targeting NE promote climbing behavior in the FST, whereas those affecting 5-HT enhance swimming behavior (Bogdanova et al., 2013). In our study, psilocybin significantly enhanced swimming behaviors supporting the known mechanism of 5-HT targeting, while the slight, though not statistically significant, increase in climbing time aligns with findings from the OFT and EPM, suggesting that psilocybin intervention may also target NE pathways. Meanwhile, anhedonia, a core feature of reward deficits, is primarily dependent upon the DA system, which plays a crucial role in predicting, anticipating and motivating the pursuit of positive awards (Belujan and Grace, 2017; Der-Avakian and Markou, 2012). The SPT takes advantage of rodents' natural preference for sucrose consumption to assess their capacity for experiencing pleasure and sensitivity to rewarding stimuli (Scheggi et al., 2018). Low dose psilocybin was able to alleviate the anhedonia observed in psychosocial stressed WIS rats, suggesting that psilocybin may present its beneficial effects through modulating DA in addition to the classic 5-HT paradigm. A limitation of

this study is the potential confounding effect of stress-induced hypo-locomotion on behavioral assessments like the EPM and FST. While psilocybin showed significant efficacy in reducing stress-induced deficits, further investigation is needed to clarify whether these effects reflect true anxiolytic and antidepressant actions or general improvements in locomotor activity.

The HPA-axis hyperactivity we observed through chronic exposure to psychosocial stress is consistent with previous work (Wang et al., 2023; Oster et al., 2017; Packard et al., 2016). This hyperactivity is believed to be linked to changes in brain reward and motivation centers, as well as alterations in neurotransmitter activity in response to prolonged exposure to stressors (Herman et al., 2016; Packard et al., 2016; Stephens and Wand, 2012; Der-Avakian et al., 2014; Levy and Tasker, 2012). As a crucial neurotrophin, sufficient BDNF expression is essential for maintaining neural plasticity and coordinating cognitive and emotional functions (Bathina and Das, 2015; Martinowich and Lu, 2008). Previous studies have demonstrated a negative correlation

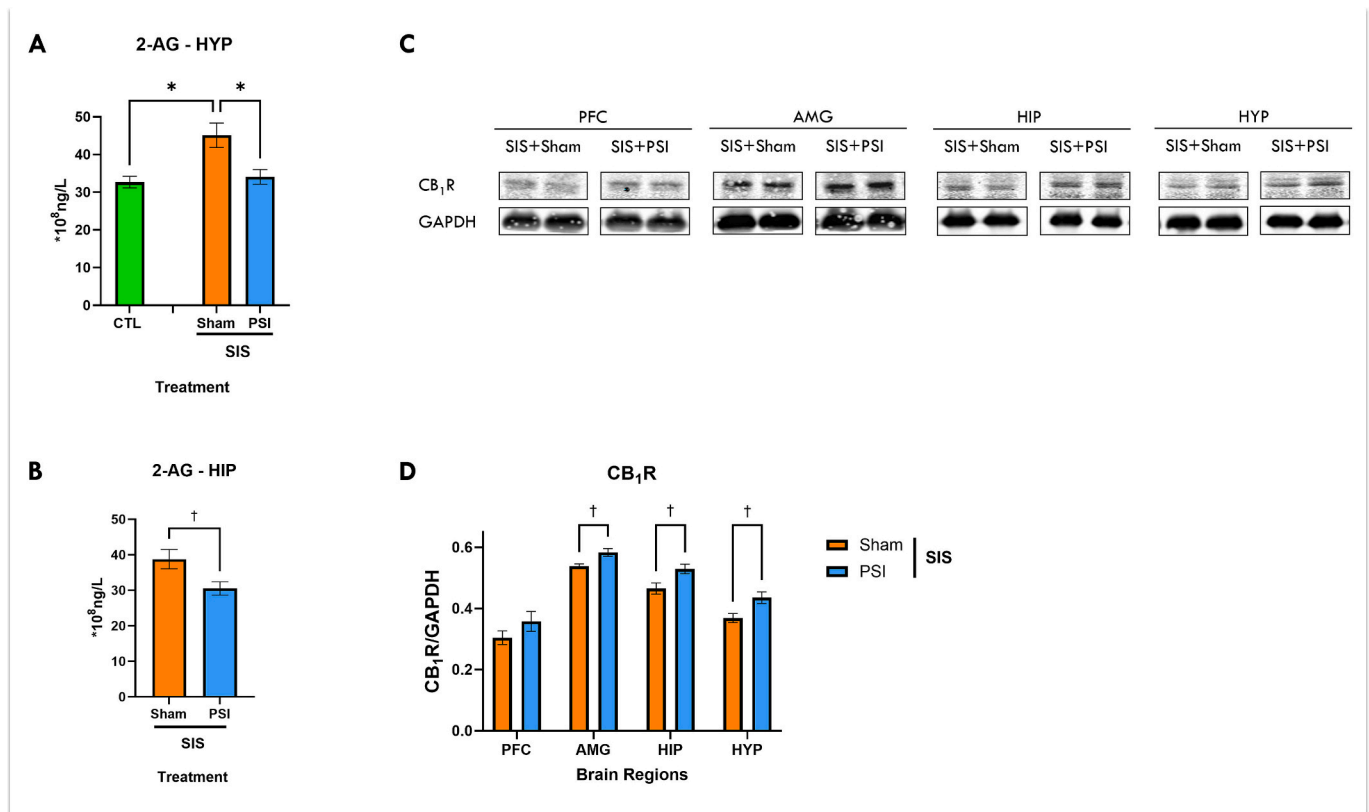


Fig. 5. Psilocybin reduces the chronic-stress induced increases of 2-AG expression while also increasing CB1R expression in four regions of the rat brain. (A) ELISA measurements of 2-arachidonoyl glycerol (2-AG) levels in the hypothalamus (HYP) revealed that SIS-Sham rats exhibited a significantly higher 2-AG level than the CTL rats, while the SIS-PSI rats had significantly reduced 2-AG level than the SIS-Sham rats. No significant differences were observed between the SIS-PSI and CTL rats. CTL $n = 3$, SIS-Sham $n = 5$, SIS-PSI $n = 5$. Data are expressed as mean $\pm$ SEM (* $p < 0.05$). One-way ANOVA followed by Tukey's post hoc test. (B) 2-AG levels, measured by ELISA, in the hippocampus (HIP), revealed that SIS-PSI rats had a significant lower level of 2-AG when compared to the SIS-Sham rats. SIS-Sham $n = 5$, SIS-PSI $n = 5$. Data are expressed as mean $\pm$ SEM († $p < 0.05$). t-test. (C) Representative western blot image showing CB₁R levels in the prefrontal cortex (PFC), amygdala (AMG), hippocampus (HIP), and hypothalamus (HYP) of SIS rats, GAPDH is used as a loading control. (D) Quantification of CB₁R levels in all four brain regions: SIS-PSI rats exhibited a significant higher CB₁R level than the SIS-Sham rats in the AMG, HIP, and HYP, while a slight but insignificant increasing was observed in the PFC compared to the SIS-Sham rats. SIS-Sham $n = 5$, SIS-PSI $n = 5$. Data are expressed as mean $\pm$ SEM († $p < 0.05$). Multiple unpaired t-test.

between CORT levels and BDNF expression, indicating that elevated CORT due to stress negatively impacts BDNF expression and neural plasticity (Mondelli et al., 2011; Schaaf et al., 2000; Nilsson et al., 2023). Our findings support these results, showing BDNF deficits in blood serum accompanied by increased CORT levels after chronic stress exposure. We found that psilocybin intervention ameliorated this decrease in BDNF levels, which aligns with a previous study that noted a single dose of psilocybin rapidly ameliorates chronic CORT exposure-induced depressive behaviors and inhibition of neuroplasticity (Zhao et al., 2024). In addition to reducing stress hormone levels, we noted that early intervention of psilocybin simultaneously restored BDNF expression levels in both blood serum and in various regions of the brain, a finding which has not been previously investigated.

Current understanding of psilocybin attributes its therapeutic potential to the agonism of 5-HT_{2A} receptor activation, which ultimately triggers the activation of various kinases, ion channels, and transporters, leading to increased neuronal firing and enhanced synaptic plasticity (Reed and Foldi, 2024). In addition, previous studies have shown that persistent psychedelic consumption can lead to 5-HT_{2A} receptor down-regulation (Pędzich et al., 2022; Urban et al., 2023). This down-regulation likely occurs due to receptor desensitization and internalization following prolonged agonist-induced activation (Mason et al., 2023), a mechanism similar to that observed with most antidepressants (Inserra et al., 2021). However, alongside the 5-HT_{2A} receptor-dependent mechanism, recent findings indicate that psilocybin may exert its antidepressant and neuroplastic effects by directly binding to

TrkB, a high-affinity BDNF receptor that is highly expressed in the HYP (Noble et al., 2011; Bathina and Das, 2015) and a primary target for antidepressants (Moliner et al., 2023). Psilocin, the active metabolite of psilocybin (Cameron et al., 2023), binds directly to TrkB with affinities 1000-fold higher than those for other antidepressants, and high-affinity TrkB positive allosteric modulators, which lack 5-HT_{2A} activity, may still retain the antidepressant potential of psychedelics (Moliner et al., 2023). Aligning with previous findings, our results indicate that psilocybin intervention significantly upregulated TrkB expression across all four brain regions, supporting the theory that psychedelics may exert their unique clinical potential by directly modulating TrkB activity, in addition to activating 5-HT_{2A} receptors (Moliner et al., 2023; Cameron et al., 2023).

Growing evidence suggests that activation of BDNF downstream signaling, especially mTOR, is required for the action of conventional and rapid-acting antidepressants (Fries et al., 2023; Athira et al., 2020), providing a novel therapeutic target for mental health condition treatment. Recent research into innovative antidepressant therapies, such as deep brain stimulation, transcranial magnetic stimulation, and ketamine, have also highlighted the critical role of mTOR as a central regulator and promising target for treating neurological disorders (Sun et al., 2022; Kato, 2023; Dębowska et al., 2023). Our results demonstrate that psilocybin intervention modulates downstream signaling cascades differently in different brain regions. The differences of mTOR signaling activation across diverse brain regions may attribute to regional variations among the four brain regions. Different brain regions exhibit

distinct patterns of neurotransmitter systems (Hansen et al., 2022) and neural circuitries (Tafet and Nemeroff, 2016; Godoy et al., 2018), and all these factors contribute to and influence how psilocybin interacts with neuronal networks. Moreover, previous studies indicate that the PFC and HYP are pivotal hubs within brain circuits crucial for higher-order cognitive functions (such as decision-making), emotion regulation, and stress response (Tafet and Nemeroff, 2016; Godoy et al., 2018; Arnsten, 2009). Given their roles heavily involved in processing stress stimulus and possessing extensive connections with other brain regions, the PFC and HYP may exhibit more pronounced changes than other brain regions under the influence of psilocybin.

The ECS mediates the fast-feedback inhibition of HPA-axis through CORT-induced synthesis of eCBs and the inhibitory properties of the CB₁R (Hill et al., 2010; Danan et al., 2021; Lu and Mackie, 2016). By binding to cannabinoid receptor type I (CB₁R) on the presynaptic terminals of glutamatergic neurons, eCBs inhibit the release of glutamate onto the CRH neurons, effectively dampening the excitatory synaptic drive, CRH neuronal activity, and downregulating the activation of the HPA-axis (Wang et al., 2023; Hill et al., 2010). The significant increase in 2-AG levels observed in the HYP of the chronic SIS rats in comparison to the CTL rats correspond to previous studies (Wang et al., 2023; Danan et al., 2021; Morena et al., 2016). This result, in collaboration with other studies (Morena et al., 2016) demonstrates that chronic homotypic stress (such as restraint and social defeat) reliably elevates 2-AG levels within the hypothalamus and hippocampus. Increased levels of 2-AG in the brain was believed to inhibit stress responses, however, chronically elevated 2-AG signaling has been shown to persistently activate CB₁R ultimately leading to CB₁R desensitization and downregulation, impaired stress response and an associated increased risk of developing mental health disorders (Morena et al., 2016; deRoos-Cassini et al., 2020). Several studies manipulating ECS function, such as CB₁R knock-out or 2-AG hydrolase deletion, have shown that impaired ECS function leads to depressive- and anxiety-like behavior, as well as cognitive decline (Soriano et al., 2021; Bedse et al., 2020).

We have demonstrated the potential effects of psilocybin on modulating the ECS for the first time. The observed 2-AG reductions suggest that early intervention with low dose psilocybin can significantly mitigate stress-induced eCB dysregulation. In addition, the dampened 2-AG levels in the HIP and HYP correlated with elevations of CB₁R in all four brain regions, suggesting that low-dose psilocybin markedly promoted CB₁R expression in all representative brain regions involved in cognitive and emotional function. However, whether psilocybin directly modulates the ECS, or if it indirectly acts *via* the HPA-axis, is unknown. Various studies have investigated the modulation of the ECS by ketamine, a dissociative anesthetic with psychedelic effects (Wolfson and Vaid, 2024), however, no studies have investigated psilocybin's impact on the ECS. Activation of 2-AG signaling has been shown to be essential for the psychostimulant and reinforcing effects of ketamine (Xu et al., 2020), whereby blockade or knock-out of the CB₁R attenuates ketamine-induced neuronal remodeling and neurobehaviors (Xu et al., 2020), as well as central antinociception (Pacheco et al., 2019). Hence, it is possible that other psychedelic substances, including psilocybin, may also directly modulate the ECS. To clarify how CB₁R activity contributes to psilocybin's antidepressant-like and cognitive effects, future studies could utilize selective CB₁R antagonists, such as Rimonabant (SR141716A) and Drinabant (AVE1625). These compounds, as demonstrated in previous studies, can provide critical insights into the role of CB₁R in mediating the therapeutic effects of psilocybin (Khan et al., 2021; Black et al., 2011).

We have shown that, in addition to its role in modulating neurotransmitters, psilocybin modulates the HPA axis and may directly, or indirectly, alter the ECS. Future studies will be needed to determine the primary role by which psilocybin acts. Moreover, previous work demonstrates that persistent psychedelic application, including psilocybin, can lead to 5-HT_{2A} receptor downregulation (Pędzich et al., 2022; Urban et al., 2023). This downregulation likely occurs due to receptor

desensitization and internalization following prolonged agonist-induced activation (Mason et al., 2023), a mechanism similar to that observed with most antidepressants (Inserra et al., 2021). These findings support the notion that psychedelics can directly affect the HPA axis, albeit from the perspective of prolonged administration. This suggests that psilocybin may directly modulate the HPA axis, leading to indirect changes in the endocannabinoid system (ECS), even though more evidence and efforts are needed.

Ethical Statement

All animal procedures strictly complied with the Guidelines of the Canadian Council on Animal Care (CCAC). All treatment and testing procedures were approved by the Animal Care and Use Committees (ACUC) of the Health Sciences Laboratory Animal Services (HSLAS) of the University of Alberta.

CRediT authorship contribution statement

Zitong Wang: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Brett Robbins:** Writing – review & editing, Project administration. **Ryan Zhuang:** Writing – review & editing, Project administration. **Thaisa Sandini:** Writing – review & editing, Methodology, Conceptualization. **Rebekah van Bruggen:** Writing – review & editing, Methodology, Conceptualization. **Xin-Min Li:** Writing – review & editing, Supervision, Funding acquisition. **Yanbo Zhang:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors have nothing to disclose.

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